

YUANRONG(REX) HAN

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Professional Summary

- AI Software Engineer with 2+ years of experience building production-oriented AI and software systems, including **RAG chatbots**, LLM-powered workflows, REST APIs, and transcription-to-note generation pipelines using **Python**, **LangChain**, **FastAPI**, **OpenAI API**, Anthropic API, and cloud infrastructure.
- Experienced in translating real user needs into practical AI solutions through **cross-functional collaboration** with professors, students, research stakeholders, and technical teams; delivered student-facing AI tools focused on accuracy, accessibility, usability, and measurable engagement improvements.
- Strong applied AI engineering background across **retrieval systems**, **prompt engineering**, **API integrations**, frontend/backend development, **Docker-based deployment**, and responsible implementation of AI systems.

Education

University of Toronto, Master of Engineering, GPA: 3.9/4.0 2024 - 2026

Major: Electrical and Computer Engineering

University of California, Berkeley, Bachelor of Science, GPA: 3.5/4.0 2018 – 2021

Major: Electrical Engineering and Computer Science

Skills

AI / LLM Tools: LangChain, Retrieval-Augmented Generation(**RAG**), Prompt Engineering, **Qdrant**

Backend Development: FastAPI, REST APIs, Jest, Supabase, Firebase

Cloud, DevOps & Infrastructure: Git, Docker, Kubernetes, DigitalOcean, Linux/Unix, AWS

Product & Development Practices: Digital Products, Prototyping, Risk Management

Languages: English, Chinese

AI Engineering Experience

AI Software Engineer, University of Toronto, Toronto, Ontario Aug 2024 – Dec 2025

- Developed a production-ready server-side retrieval-augmented chatbot (**RAG**) using **Python (FastAPI)** and JavaScript, integrating machine learning pipelines to ensure real-time, contextually accurate responses.
- Implemented server-side response streaming so users saw output incrementally instead of waiting for full generation, cutting time-to-first-token and improving perceived responsiveness.
- Containerized and deployed the service with Docker, establishing a repeatable deployment workflow that moved a research prototype into a real, student-facing tool.

Graduate Research Assistant, University of Toronto, Toronto, Ontario Oct 2024 – July 2025

- Designed and prototyped an AI-powered lecture note-taking system using **Azure AI Studio** to improve accessibility for students unable to take notes.
- Integrated **Whisper** for speech-to-text transcription and connected transcript outputs to LLM-based pipelines for automated note generation.
- Partnered with researchers to translate an accessibility research goal into a working prototype on Azure AI Studio, supporting students unable to take notes.
- Developed structured prompting workflows to convert lecture audio into organized summaries, key points, and student-facing study notes.

Software Engineering Experience

Mobile Software Engineering, Dream Formula Education, San Francisco, California May 2021 – June 2022

- Led the development of a Swift based iOS app, boosting academic task organization for students and driving a 30% increase in user engagement.
- Integrated secure **Firebase** authentication and database solutions to facilitate seamless onboarding for new users while ensuring data integrity.
- Employed **CI/CD** practices and **Git** for version control, streamlining deployment processes and enabling consistent updates informed by pilot user feedback, which significantly enhanced app performance and supported predictive models for future feature forecasting.